

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 28

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B - Steam washing (hot water / steam jets kill organisms in the sediment) (1)	1
01.2	<i>1 mark for each point.</i> • Chemicals cause the oil molecules to join together / solidify (1) • Producing more solid material that can be collected / lifted more easily than liquid oil (1)	2
01.3	<i>1 mark for each point.</i> • Air contains about 210 cm ³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm ³ per litre (less than 3% of the amount in air) (1) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off (1) • Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff (1) • Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest (1) • Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture (1) • Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field; slows runoff (1)	5
02.2	<i>1 mark for each point.</i> • No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles (1) • Permanent root systems and canopy cover protect the soil all year (1)	2
02.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Control stocking density so vegetation regrows faster than it is eaten (1) • Restrict access to vulnerable areas such as river banks (fencing) (1) • Move / remove livestock at high-risk times, eg after heavy rain when soil is wet (1) • Rotational grazing to rest pastures (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
03.1	<i>1 mark for each point.</i> • Cost increases as the percentage prevented increases (1) • The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90% (1)	2
03.2	<i>1 mark for the correct answer.</i> • $16.0 - 5.0 = 11.0$ (£ million) (1)	1
03.3	<i>1 mark for each point.</i> • ALARA: emissions should be As Low As Reasonably Achievable (1) • It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set (1) • BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive (1) • The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% (1)	4

Question	Marking guidance	Marks
03.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen (1) • Better understanding of the health / environmental effects so stricter standards are set (1) • Improved technology becomes available (BATNEEC changes) / cost of better control falls (1) • Pollutant may have accumulated (persistent) so background levels are higher (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
04.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Unburnt fuel in vehicle exhaust gases (1) • Evaporation from fuel tanks / spillage at filling stations (1) • Solvents (paints, cleaning products) (1) • Aerosol propellants (1) • Gas leaks / fossil fuel exploitation (methane) (1)	3
04.2	<i>1 mark for each point.</i> • Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank (1) • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon (1)	2
04.3	<i>1 mark for each point.</i> • A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe (1) • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) (1) • NO_x are chemically reduced to nitrogen and oxygen (1)	3
04.4	<i>1 mark for each point.</i> • Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues (1) • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	
05.1	<i>1 mark for each point.</i> • Vehicles: high energy density (200 Wh kg⁻¹) minimises battery mass so range is greater; many cycles and high efficiency (1) • Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable (1) • Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management (1) • Nickel-cadmium largely replaced because of cadmium toxicity and cost (1)	4
05.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Recharging speed (1) • Self-discharge during non-use (1) • Availability of raw materials (lithium, cobalt) (1) • Recyclability / disposal (1)	2

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> • Parked electric vehicles plugged into the grid form a very large distributed battery (1) • At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines (1)	2
05.4	<i>1 mark for each point.</i> • Surplus electricity electrolyses water to hydrogen, which can be converted to methane (1) • Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	
06.1	<i>1 mark for each point.</i> • Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated (1) • Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time (1) • Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it (1) • Repeat each treatment several times and calculate means (and standard deviation) (1) • Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance (1)	5
06.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Rainfall intensity / raindrop size / height (1) • Soil texture, compaction, organic matter, depth (1) • Permeability of the material beneath (1) • Gradient (1) • Root binding (1)	3
06.3	<i>1 mark for each point.</i> • All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead (1) • Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	
07.1	<i>1 mark for each point.</i> • Friction with the prevailing south-westerly winds drags surface water north-eastwards from the tropical Atlantic (1) • Water in the NE Atlantic cools, becomes denser (and more saline as ice forms) and sinks, drawing warm surface water in to replace it; the cold water returns southwards at depth (thermohaline circulation) (1)	2

Question	Marking guidance	Marks
07.2	<i>1 mark for each point.</i> - Higher temperatures melt land ice on Greenland which flows into the North Atlantic (1) - The fresh water dilutes the seawater, reducing its salinity and density (1) - Less dense water sinks less easily so the flow rate of the current is reduced (1) - Less heat is transported to NW Europe so the UK could become colder (like Moscow / Churchill at the same latitude) even while global temperatures rise (1)	4
07.3	<i>1 mark for each point.</i> - Several interconnected systems act at once over different timescales: jet stream changes may raise temperatures, a slowing Conveyor may lower them, greenhouse gases raise them (1) - Natural fluctuations (solar output, orbit) can hide or exaggerate anthropogenic change; individual events do not show trends (1) - Time delays between cause and effect (oceans take a long time to warm because of volume and heat capacity); incomplete understanding of the processes (1)	3
07.4	<i>1 mark.</i> Argo floats sink to a set depth for about 10 days, then surface and transmit temperature, salinity and position data; the sequence of positions shows current direction and speed (surface currents also by satellites and buoys) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-97; AQA spec 3.2.1.2, 3.2.2.3.	
08.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Aerodynamic shapes / fairings reduce friction with the air (air resistance) so less energy is needed to propel the vehicle, especially at speed; hydrodynamics for ships (2) - Reduced mass (plastics, carbon-fibre composites, high-strength steel alloys, aluminium engine blocks, lighter batteries and magnets, rounded shapes) means less energy to accelerate and less rolling resistance, so lower fuel consumption (2) - Conventional friction brakes convert kinetic energy to waste heat; regenerative braking uses a generator to convert it to electricity stored in a battery (or returned to the rail grid) which then propels the vehicle, so less fuel is needed to accelerate again (2)	6
08.2	<i>1 mark for each point.</i> - Pneumatic tyres deform as they rotate; the continual change in shape and air movement inside generates frictional heat (more if under-inflated) (1) - Steel wheels do not deform so rolling resistance is very low (1)	2
08.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - More valves per cylinder to remove exhaust gases fully so they do not mix with fresh fuel (1) - Better temperature control (thermostatic fan) (1) - Electronic ignition control / regular servicing to keep spark timing correct (1) - Turbochargers supplying more oxygen (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 225-227; AQA spec 3.3.4.2.1.	

Question	Marking guidance	Marks
09.1	<i>Correct answer only.</i> • B - Tropospheric ozone (formed by reactions of NO _x in sunlight) (1)	1
09.2	<i>1 mark for each point.</i> • Primary pollutant: released directly by human activities, eg sulfur dioxide from burning coal / smelting sulfide ores; NO _x from high-temperature combustion (1) • Secondary pollutant: produced by chemical reactions between primary pollutants, often with non-pollutants (water, oxygen, sunlight) (1) • Sulfur dioxide is oxidised by ozone to sulfur trioxide which dissolves to form sulfuric acid; NO _x dissolve to form nitric acid (1) • The acids (sulfurous, sulfuric, nitric) are secondary pollutants; ozone itself is a secondary pollutant of NO _x photochemistry (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 237-238; AQA spec 3.4.1.	
10.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • No reservoir range = $3.9 - 0.7 = 3.2$ m; with reservoir = $2.7 - 1.7 = 1.0$ m (1) • Reduction = 2.2 m (1)	2
10.2	<i>1 mark for each point.</i> • After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream (1) • During dry periods the dam is opened to release water and maintain river levels (1)	2
10.3	<i>1 mark for each point.</i> • Deforestation / urbanisation reduce infiltration and interception (1) • So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Site factors: topography - narrow exit from a large deep basin so a small dam holds a large volume; impermeable, strong bedrock without faults or seismic activity; large catchment with regular high rainfall and low evaporation; catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation; forest / vegetation in the flooded area decaying anaerobically releasing methane; access for workers and materials and distance to demand; value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • Alternatives for supply: rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • Conservation strategies: low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • Evaluation: reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination has high energy / carbon cost; combination needed as demand grows • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

Question	Marking guidance	Marks
11.1	Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Why non-renewable: formed too slowly; geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock; Lasky's principle; cut-off ore grade and its drivers • Extraction technologies: mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • Reducing demand: recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • Environmental limits: impacts of mining and their control; energy costs rising with lower grades; conflicts • Judgement: technology can extend supplies substantially but only within energy and environmental constraints; circular economy essential Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25